

Chapter 6

Cost Accounting and Analysis

Cost accounting refers to the classification of financial information from a budget or financial reports into cost categories based on the agency's cost structure analysis regarding the relationship of resources spent and major services provided. The [Oregon Accounting Manual 2010](#) of the Department of Administrative Services defines cost accounting as follows.

A method of accounting which provides for assembling and recording all elements of cost incurred to accomplish a purpose, to carry on an activity or operation, or to complete a unit of work or specific job.

Cost analysis is the use of cost data to analyze the determinants of costs and evaluate changes in agency operation. It is the basis for setting user fees for public services and is a decision tool for service contraction or expansion. This chapter covers aspects of cost accounting and analysis. There are [slides](#) as well as three YouTube videos associated with this chapter:

- General concepts are outlined in the video [Cost Accounting Concepts](#)
- The [Hoosier County Road Department video](#) is an example of a non-personnel cost analysis.
- The [Hoosier Police Department video](#) is an example of a personnel cost analysis.

Note that the latter two videos are linked again in the respective sections below. We also cover break-even analysis using garbage collection as an example.

Before diving into the topics, let us define some terminology first:

- **Obligation:** An obligation is the responsibility to meet the terms of a contract. For example, road salt purchases in June for delivery in October

- **Outlay:** Outlays are the physical outflow of cash. For example, the payment for road salt in September.
- **Cost:** Actual use and consumption of the resource. For example, road salt consumption in December and January as cost in two different calendar years

Usually, there are small dollar amount differences between obligations, outlays, and cost.

6.1 Cost Accounting

Cost accounting aims to assign costs to services and categories in a manner that supports effective cost analysis. This process relies on various data sources, including past budgets, operational reports, and contracts or billing statements. A key aspect of cost accounting is the determination of mission or service centers, which involves categorizing the primary services provided to the public. For example, in a Department of Transportation, road maintenance could be identified as a key service center. This categorization helps to accurately track and analyze costs associated with specific functions or services.

In cost accounting, direct costs refer to expenses that can be directly and accurately assigned to a specific service or project. For example, the cost of asphalt used to repave Meridian Street in Indianapolis is a direct cost because it is clearly linked to that specific project. On the other hand, indirect costs are those that cannot be precisely assigned to a specific service due to the lack of a direct causal link between the expense and the service provided. An example of an indirect cost would be the salary of a manager overseeing road maintenance in Indianapolis, as their work supports multiple projects rather than a single, specific service. One of the goals of cost accounting is to convert as many indirect costs as possible into direct costs by creating more precise links between costs and the services provided, improving the accuracy of cost allocation.

Cost concepts in accounting help distinguish how expenses behave in relation to the level of output or services provided:

- **Fixed cost:** This type of cost remains constant and does not vary with the level of output. An example is office space rent, which remains the same regardless of how much work or service is being performed.
- **Variable cost:** These costs fluctuate based on the level of goods or services provided. For instance, the cost of paving roads using asphalt increases as more roads are paved.
- **Semi-variable cost:** These costs contain both fixed and variable components. An example is a copy machine, where there is a fixed rental fee but additional costs accrue based on the number of copies made.
- **Step cost:** These costs change with output but in discrete jumps rather

than continuously. For example, a school may need to hire one additional teacher for every 20 students, resulting in a step-like increase in costs as student numbers grow.

Other key cost concepts include the full (total) cost which includes all resources utilized to produce a service. The unit cost is the total cost divided by total quantity. In economics, this is called the average total cost. For a government agency (but also for industry and nonprofits) it is important to know the average total cost. If the price per unit sold (or service provided) is below the average total cost, the service provision is not sustainable in the long run. Finally, we have marginal cost which is the change in total cost if one more unit of service is provided or one more unit of output is produced.

6.2 Cost Accounting Steps

The cost accounting involves the following steps to assign costs to specific services or mission centers:

1. Identification of mission centers
2. Identification of activity and output units
3. Identification of expenditure categories
4. Identification of allocation factors
5. Linking cost to mission centers

In the following, we are going to describe each of those steps in detail.

Identification of Mission Centers: This first step involves categorizing the parts, i.e., mission centers, of the organization that provide direct services to the public. These mission centers represent separate services for which unit costs need to be developed. For example, a water department may have the mission centers “water distribution” and “water treatment.” Other examples would be “fire suppression” and “fire prevention” for a fire department or “garbage collection” and “garbage disposal” for a sanitation Department. By identifying these mission centers, organizations can better assess the cost of providing each service and appropriately allocate resources or set rates for users.

Identification of Activity and Output Units: This entails defining measurable units of service or output (e.g., miles of road maintained, number of public safety incidents handled) to calculate per-unit cost of services. Each mission center needs to have clearly defined, measurable units of activity or output to allow for accurate cost analysis. These units should directly reflect the service provided by the mission center. Identifying these measurable activity or output units allows organizations to allocate costs effectively and calculate the cost per unit of service provided. For example:

- In a nursing home, the output unit might be the number of resident days.
- For garbage collection, it could be the number of tons of waste collected or the number of pickups.

- In a water department, the unit might be the volume of water treated or distributed, measured in gallons or liters.
- For a police department, the unit could be the number of incidents responded to or cases handled.

Identification of Expenditure Categories: Costs are categorized into groups such as labor, materials, or overhead to facilitate better tracking and analysis. Usually, we differentiate between personnel and non-personnel expenditures. Personnel expenditures are usually composed of salaries and fringe benefits (e.g., health insurance, Social Security). Non-personnel expenditures include items such as supplies, equipment, utilities, or facilities

Identification of Allocation Factors: Allocation factors are determined to fairly distribute indirect costs (like administrative expenses) across different mission centers based on relevant drivers, such as hours worked or resources used. Requirement of using allocation factors to link indirect cost to mission centers

Indirect Cost	Allocation Factor
Facilities	Square footage
Overhead	Direct personnel or direct cost
Liability insurance	Direct personnel or output
Computer usage	Direct personnel or output
Printing services	Output

Linking Costs to Mission Centers: In a fifth step, costs are linked to the mission centers based on direct and allocated expenses, providing a clear picture of how resources are consumed by each service or function. In the previous section, we differentiated between direct cost (i.e., expenses that are directly assigned to a specific mission center) and indirect costs (i.e., expenses that cannot be directly assigned to a specific mission center). There is also overhead which refers to costs associated with providing administrative services to all mission centers. This includes the salaries of management, as well as insurance, utilities, and facilities costs, particularly when usage cannot be directly linked to a specific mission center.

6.3 Hoosier County Road Department

Let us consider the Hoosier County Road Department whose mission is to maintain the county's roads (see [Hoosier County Road Department video](#)). All personnel work full time and 225 days per year. The personnel maintaining the roads cost \$74,000 per employee per year. That amount includes the salary and fringe benefits. There are two types of roads in the county, i.e., gravel roads (80 miles) and paved roads (185 miles). The state reimburses the county with \$6,250 and \$10,200 per mile of gravel road and paved road, maintained. Below is a summary of the activity and cost information that was compiled.

	Gravel	Paved	Total
Supplies expenditures	\$41,000	\$180,000	\$221,000
Miles Repaired	80	185	265
Cost per Mile	\$513	\$973	
Personnel (FTE)	4	5	9
State reimbursement per mile	\$6,250	\$10,200	
Productivity (miles/person/year)	20	37	
Maintenance employee salary	\$74,000	\$74,000	
Director's salary			\$96,000
Assistant's salary			\$61,000
Equipment lease: Rate/month	\$6,250	\$9,750	
Equipment lease: Rate/mile	\$725	\$1,100	
Maintenance facility			\$210,000

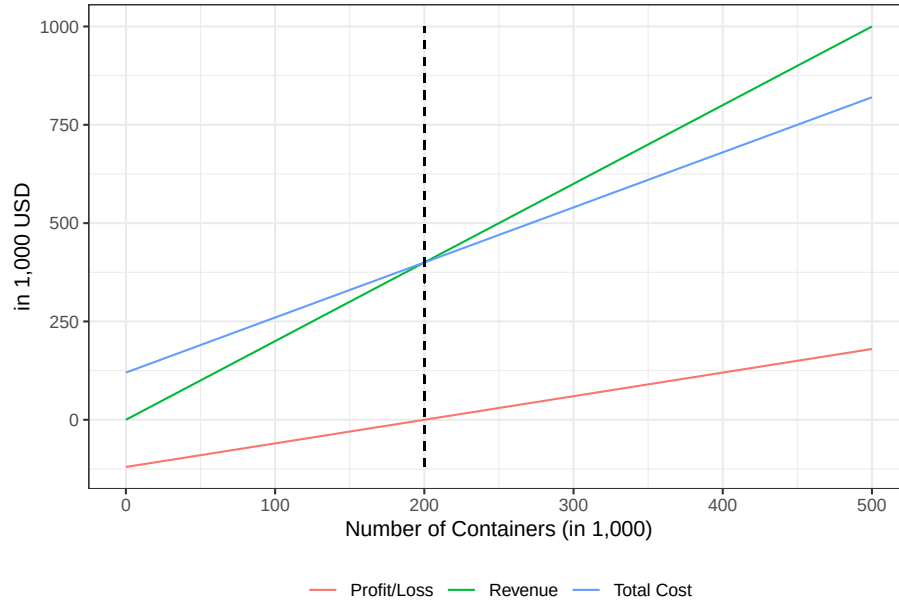
6.4 Break-Even Analysis

The city department for municipal solid waste (MSW) collection serves about 250,000 households per year who all own one garbage container. The collection fee is \$2 per 96 gallon container. Thus, the revenue in this example would be \$500,000. On the cost side, we have annual fixed cost that are composed of administrative services (\$35,000) and equipment lease (\$85,000). The variable cost per container is composed of a landfill charge (\$1) and equipment operation on collection routes (\$0.40). Suppose that due to inflation, the landfill considers increasing the fee from \$1.00 to \$1.30 per container. The table below calculates the total revenue, fixed cost, variable cost, and total cost and the old and proposed landfill (LF) fee.

Item	Current LF Charge	New LF Charge
Number of containers	250,000	250,000
Collection Fee	\$2	\$2
Revenue	\$500,000	\$500,000
Administration	\$35,000	\$35,000
Equipment Lease	\$85,000	\$85,000
Fixed Cost	\$120,000	\$120,000
Landfill charge	\$1.00	\$1.30
Equipment operating on collection routes	\$0.40	\$0.40
Total variable costs per unit	\$1.40	\$1.70
Variable Cost	\$350,000	\$425,000
Total Cost	\$470,000	\$545,000
Surplus/deficit	\$30,000	\$(45,000)

The graph below represents the break even number of containers under the old

landfill charge.



6.5 Personnel Cost versus Non-Personnel Cost

We make the distinction between personnel and non-personnel cost. Personnel cost are usually the largest single component of a budget and include wages, salaries, and fringe benefits. In general, total salary expenses are defined as $N \times w$ where N represents the number of workers in a given working grade and w represents the wage per worker for said working grade. Whereas non-personnel cost are relatively straightforward to calculate and ultimately result in the cost per unit produced or provided times the number of units, personnel cost need to keep track of the steps (grades) which vary with education, experience, and tenure of each employee.

Budgeting for non-personnel cost can be done by the so-called incremental method, which adjust past cost for inflation, i.e., last years expenditures times $(1 + \text{inflation rate})$. Since this method does not account for the possibility of input substitution, the unit cost method can be use that calculates \$ per unit of output (unit costs).

6.6 Personnel Budget

To illustrate the preparation of a personnel budget for a police department (PD), assume that you are responsible for the budget construction of Hoosier PD (see

[Hoosier Police Department video](#)). Given the population size of the district, you need nine patrol cars at any given time. Patrols are 24 hours per day and 365 days per year. For safety reasons, each patrol requires two officers per car. Those officers work eight hour shifts per day on five days per week. All officers receive three weeks of vacation and and sick leave (i.e., two weeks of vacation and one week of sick leave). The salary per police officer is \$61,200 with 30\$ fringe benefits. The estimation of the personnel budget requires four steps:

1. **Calculation of personnel hours needed:** 24 hours per day $\times$ 365 days per year $\times$ 2 officers per car $\times$ 9 patrols = 157,680 hours
2. **Calculation of provided hours per officer:** One officer works 49 weeks (52 minus three weeks) and eight hours per day. Given that the officer works for five days per week, the number provided is 49 weeks $\times$ 5 days per week $\times$ 8 hours per day = 1,960 hours
3. **Required number of full-time officers:** 157,680 divided by 1,960 = 80.45. Hence, the budget needs to be done of 81 police officers. Given that the service level is required and no overtime is assumed, the number of police officers is rounded up.
4. **Salary and fringe benefit costs:** Given the salary of \$61,200, the fringe benefits are an additional \$18,360. Hence the total budget is $81 \times \$79,560 = \$6,444,360$.

6.7 Exercises

1. **MCPTA Break-Even Analysis (***)**: The Metro City Public Transportation Authority (MCPTA) is considering expanding its fleet from currently 150 to 200 buses to accommodate growing demand. Each bus has a capacity of 60 passengers. The average operating cost per bus is \$450 per day and includes fuel, maintenance, depreciation, and the driver's salary. MCPTA charges a flat fare of \$1.50 per ride. Each bus does 10 round trips per day on its route and operates 365 days a year. The buses currently operate at 70% capacity on average. Overhead is currently \$4 million. MCPTA expects that the new expansion would decrease ridership because of the transport network becoming larger. In addition, fixed costs (e.g., infrastructure, administrative overhead) would rise by an additional \$500,000 annually with this expansion.
 - (a) Calculate and report MCPTA's profit without the expansion. Assuming the load capacity dropping to 65% after the expansion, what is the new profit with 200 buses?
 - (b) Calculate the break-even capacity in percent that MCPTA needs in order to cover the new total operating costs with the expanded fleet. Show all steps in your calculations, including the new total costs, revenues, and break-even analysis.
 - (c) If MCPTA considers raising the fare to \$1.75 per ride, how would this affect the break-even number of daily passengers? Would this

new fare reduce the financial risk of the expansion? Explain.

- (d) Discuss other public finance factors that could impact MCPTA's decision to expand (e.g., subsidies, long-term infrastructure costs, elasticity of demand, social welfare, or environmental benefits). What are your recommendations?

2. ***Municipal Electric Utility Expansion*** (***) : The City of Hoosier Valley operates its own municipal electric utility (HV Electric) and provides electricity to 100,000 households. The current generation capacity is 500 gigawatt-hours (GWh) per year, and the utility operates at 85% of its capacity. The utility charges \$0.13 per kilowatt-hour (kWh), and the total annual cost (including fixed and variable costs) is \$40 million. The fixed costs, which include infrastructure maintenance, administrative overhead, and long-term debt servicing, account for 60% of the total annual cost. Due to population growth and increasing energy consumption, the city anticipates a 20% increase in demand for electricity over the next 5 years given current electricity prices. HV Electric is considering expanding its capacity to 600 GWh by investing in renewable energy sources (solar and wind). The expansion would increase the annual fixed cost 10 million in one-time capital costs, spread equally over the next 5 years, plus an ongoing annual increase of \$2 million in variable operating costs (labor and maintenance). The city is also considering raising the electricity rate to \$0.15 per kWh to finance the expansion and offset rising costs. The proposed rate increase is expected to reduce overall electricity usage such that the utility load decreases by 3 percentage points to 82%.

- (a) Assuming no expansion, calculate and report the current cost per kWh for delivered electricity and the current annual profit of HV Electric.
- (b) Given the expansion and the proposed rate increase, calculate and report the new cost per kWh for delivered electricity and the new annual profit of HV Electric.
- (c) Calculate the break-even point for HV Electric's the expansion capacity scenario in terms of load capacity. That is, the 3 percentage point drop may be an underestimation and the board of directors of HV Electric would like to know the maximum load capacity drop that allows them to remain profitable.
- (d) Opponents to the rate increase claim that even without any rate increase (and thus, without drop in demand), HV Electric is profitable. Is this claim correct? Support your answer with the relevant calculations.